ZHENGHANG XU

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RESEARCH INTERESTS

AI for Queueing Discovery and Control
I develop AI and learning methods to understand queueing systems from data, evaluate how policy changes affect their performance, and improve decisions such as capacity allocation and pricing.

Methodologies	Causal Machine Learning, Queueing Theory, Optimal Control, Online Learning
Applications	Healthcare Management, Online Platforms, Inventory Management

EMPLOYMENT

Rotman School of Management, University of Toronto *2026 - Present*
Postdoctoral Research Fellow
Advisors: Nasser Barjesteh, Sheng Liu, Vahid Roshanaei

EDUCATION

Rotman School of Management, University of Toronto *2021 - 2026*
Ph.D. in Operations Management and Statistics
Advisors: Opher Baron, Dmitry Krass, Philipp Afèche

The Chinese University of Hong Kong, Shenzhen (CUHKSZ) *2017 - 2021*
B.Sc. in Statistical Science
Rank: 1/151

PUBLICATIONS AND PAPERS UNDER REVISION

Queueing Causal Models: Comparative Analytics in Queueing Systems
with Opher Baron, Dmitry Krass, Arik Senderovich and Mark van der Laan
M&SOM, 2025, 28(2):517–536 [Journal] [SSRN 5104446]
- Next-Gen Scholar's Symposium, 2026
- **First Prize,** CORS 2025 Queueing and Applied Probability SIG
- **Winner,** 2024 Oded Berman Student Paper Competition
- **Finalist,** CORS 2024 Best Student Paper Competition (Open Category)
- Service Management SIG, M&SOM 2024

Managing Scarce MRI Capacity in Overloaded Queueing Systems
with Andre Cire, Adam Diamant, Eugene Furman and Opher Baron
Major revision at M&SOM, 2026 [SSRN 6158086]
- **Finalist,** CORS 2026 Healthcare SIG

An Investigation of Treatment Barriers for End-Stage Kidney Disease Patients Using Advanced Analytics
with Olga Bountali, Sila Cetinkaya, Michael Hahsler, Farnaz Nourbakhsh and Henry Quinones
Healthcare Analytics, 2025, 8:100438 [Journal] [SSRN 5066305]

A High-Fidelity, Machine-Learning Enhanced Queueing Network Simulation Model for Hospital Ultrasound Operations
with Yihan Pan, Jin Guang, Chengwenjian Wang, Xuanming Zhang, Xinyun Chen, J.G. Dai, Jingjing Sun, Yichuan Ding, Pengyi Shi, Hongxin Pan, Kai Yang and Song Wu
In 2021 *Winter Simulation Conference (WSC)* (pp. 1–12). IEEE.

WORKING PAPERS

Learning to Price in Queues: Arrival Observability and Value Randomness
with Philipp Afèche, Opher Baron and Dmitry Krass
To be submitted, draft available upon request
- 2024 Rotman TD MDAL Research Grant

Structural Causal Inventory Models: Identification and Policy Coverage
with Opher Baron and Dmitry Krass
Work in progress

Queueing Causal Models for Emergency Department Efficiency
with Opher Baron, Dmitry Krass, Mark van der Laan and Yufeng Zheng
Work in progress
- 2025 Rotman TD MDAL Research Grant

Fairness in Multi-Stage Healthcare Access: A Study of Congestion, Chronic Patients, and Operational Redesign
with Olga Bountali, Sila Cetinkaya, Michael Hahsler, Farnaz Nourbakhsh and Henry Quinones
Work in progress

HONORS AND AWARDS

Finalist, CORS Healthcare SIG *2026*
First Prize, CORS Queueing and Applied Probability SIG *2025*
Winner, Oded Berman Student Paper Competition *2024*
Finalist, CORS Best Student Paper Competition (Open Category) *2024*
Rotman TD MDAL Research Grant ($15K total over three years) *2023 - 2025*
Ph.D. Fellowship, Rotman School of Management *2021 - 2026*
Presidential Award for Outstanding Students (10/900+), CUHKSZ *2021*
Undergraduate Research Assistantship, CUHKSZ *2019*
First Class Academic Scholarship (top 3%), CUHKSZ *2019, 2020*
Third Class Academic Scholarship (top 8%), CUHKSZ *2018*
Dean's List, CUHKSZ *2018, 2019, 2020*

PROFESSIONAL SERVICE

Session Chair, CORS *2025*
Chaired a session of the Queueing and Applied Probability SIG.
Co-organizer, Rotman Young Scholar Seminar *2025 - Present*
Co-organize weekly seminars and coordinate speaker invitations.
Journal Referee for *Operations Research, Queueing Systems, INFOR.*
Conference Referee for *M&SOM Main Conference.*

INVITED TALKS & CONFERENCE PRESENTATIONS

Queueing Causal Models: Comparative Analytics in Queueing Systems

* denotes presentations I gave jointly with another speaker.

Invited Seminars

- Next-Gen Scholar's Symposium, National University of Singapore — *Aug 2026*
- School of Economics and Management, Tsinghua University — *Dec 2025*
- College of Business, Shanghai University of Finance and Economics — *Dec 2025*
- Antai College of Economics and Management, Shanghai Jiao Tong University — *Dec 2025*
- College of Business, City University of Hong Kong — *Dec 2025*
- Institute of Supply Chain Analytics, Dongbei University of Finance and Economics — *Dec 2025*
- School of Management, Fudan University* — *Sep 2024*
- Antai College of Economics and Management, Shanghai Jiao Tong University* — *Sep 2024*
- School of Business, Renmin University of China* — *Sep 2024*
- School of Management, Hefei University of Technology* — *Sep 2024*
- School of Management and Economics, The Chinese University of Hong Kong, Shenzhen* — *Sep 2024*
- Department of Decisions, Operations and Technology, CUHK Business School* — *Sep 2024*
- IEDA, Hong Kong University of Science and Technology* — *Sep 2024*
- Department of Logistics and Maritime Studies, The Hong Kong Polytechnic University* — *Sep 2024*

Conference Presentations

- INFORMS Applied Probability Society — *Jul 2025*
- Matrix-Analytic Methods (MAM) 12 Conference (Keynote)* — *Jun 2025*
- CORS Queueing and Applied Probability SIG — *Jun 2025*
- INFORMS Annual Meeting — *Oct 2024*
- CORS Student Paper Competition (Open Category) — *Jun 2024*
- Rotman TD MDAL Research Round Table — *May 2024*
- INFORMS Annual Meeting — *Oct 2023*
- CORS Annual Meeting — *Jun 2023*

Learning to Price in Queues: Arrival Observability and Value Randomness

- Rotman TD MDAL Research Round Table — *May 2026*
- POMS-HK Conference — *Jan 2026*
- INFORMS Annual Meeting — *Oct 2025*
- CanQueue workshop — *Aug 2025*
- Prof. Huan Zheng's Group, Shanghai Jiao Tong University (Online) — *Jul 2025*
- CORS Annual Meeting — *Jun 2025*
- INFORMS Toronto Student Chapter Research Talk Series — *May 2025*

Managing Scarce MRI Capacity in Overloaded Queueing Systems

- MSOM Conference — *Jul 2026*
- CORS Healthcare SIG — *Jun 2026*

Queueing Causal Models for Emergency Department Efficiency

- Rotman TD MDAL Research Round Table — *May 2025*

TEACHING EXPERIENCE

Guest Speaker, Rotman School of Management, University of Toronto
RSM3090: Queueing Theory - Models and Analysis *Fall 2025*

Teaching Assistant, Rotman School of Management, University of Toronto
RSM1240: Statistics for Management (MBA) *Winter 2024, 2025*
RSM1282: Operations Management (MBA) *Winter 2025*
RSM8414: Tools for Probabilistic Models and Prescriptive Analytics (MMA) *Fall 2024*
RSM8432: Management Analytics Practicum (MMA) *Winter 2024*
RSM2408: Modeling and Optimization for Decision Making (MMA) *Fall 2023*
RSM8415: Service Analytics for Management Analysis (MMA) *Winter 2023*
RSM2405: Supply Chain Management (MBA) *Fall 2022*

Project Coach, Master of Management Analytics (MMA) Program, Rotman School of Management, University of Toronto
Host companies: Imperial Oil, Ontario Health, Scotiabank *Winter 2024*

Teaching Assistant, CUHKSZ
STA2002: Probability and Statistics II (Undergraduate) *Spring 2021*
MAT2040: Linear Algebra (Undergraduate) *Spring 2020*
CHM1001: General Chemistry (Undergraduate) *Fall 2019*
BIO2001: General Biology (Undergraduate) *Spring 2019*

REFERENCES

Prof. Opher Baron
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Rotman School of Management
University of Toronto
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Prof. Dmitry Krass
Operations Management & Statistics
Rotman School of Management
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Email: dmitry.krass@rotman.utoronto.ca
Prof. Philipp Afèche
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